



KG-1 BOOK A

Abacus Mental Arithmetic

Student Information

Name:

Level:

Date:

3

5

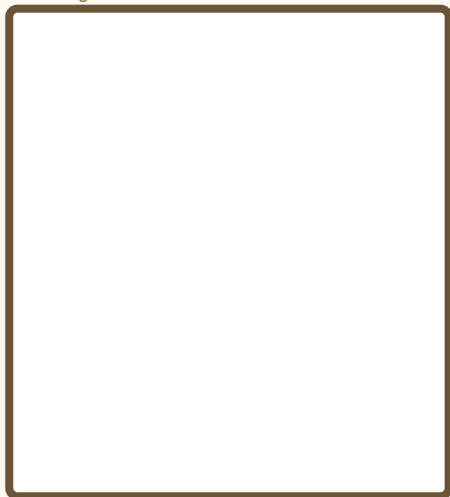


4

Description of the Abacus

Frame

The outer rectangle



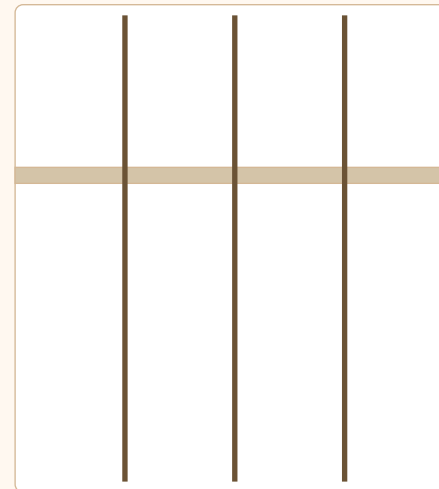
Beam

Horizontal divider bar



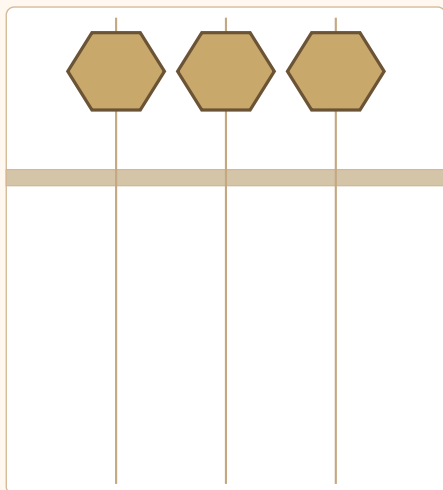
Rod

Vertical bars for beads



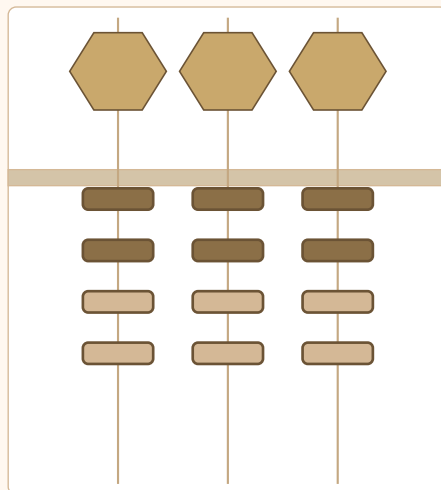
Upper Bead

Value = 5



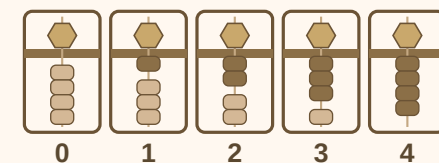
Lower Bead

Value = 1 each



Complete Abacus

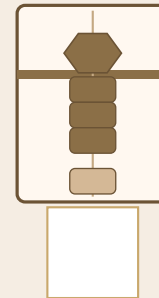
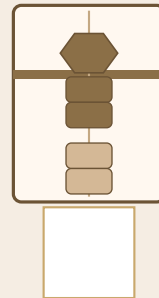
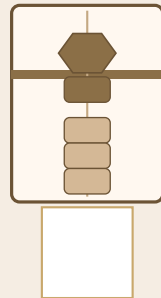
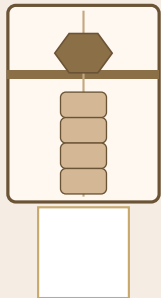
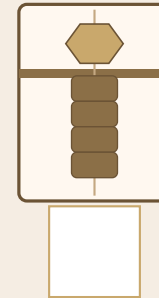
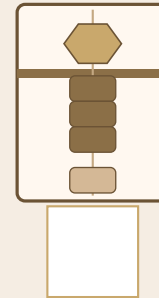
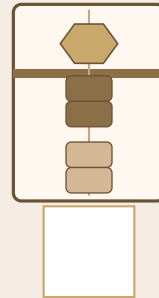
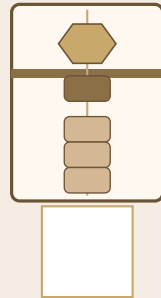
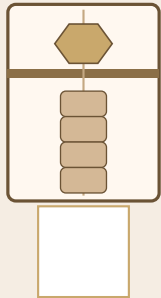
All parts together: values 0 to 9



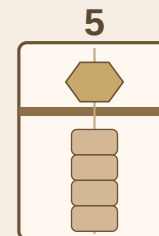
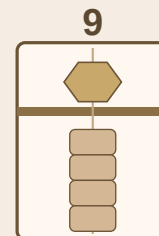
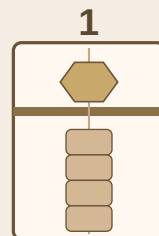
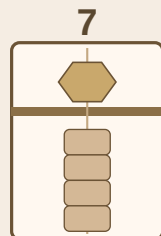
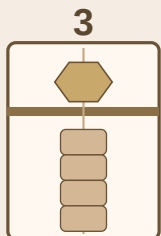
Write the Value of Beads

Set 1 to 9 on the abacus

Look at each abacus and write the number it shows in the box below.



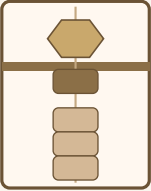
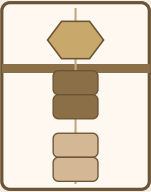
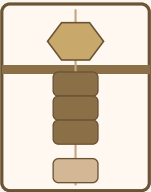
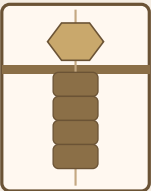
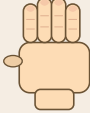
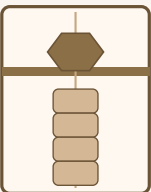
Now set these numbers on the abacus:



Understand Numbers

Numbers 1 to 5



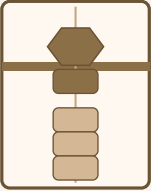
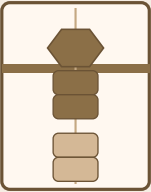
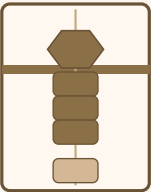
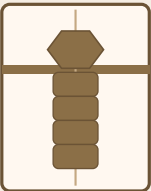
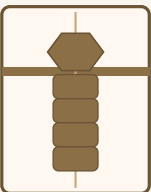
Abacus	Hand	Dots	Number	Trace	Trace	Practice
			1	1	1	
			2	2	2	
			3	3	3	
			4	4	4	
			5	5	5	



Understand Numbers

Numbers 6 to 10



Abacus	Hand	Dots	Number	Trace	Trace	Practice
	 5+1		6	6	6	
	 5+2		7	7	7	
	 5+3		8	8	8	
	 5+4		9	9	9	
	 5+5		10	10	10	



Write the Numbers You Can Identify

How many numbers can you find hidden in the picture? Write them in the boxes!



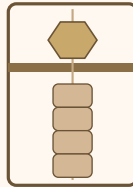
Write the numbers you found:

Count and Write

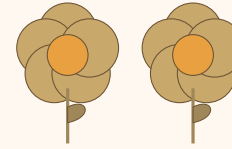
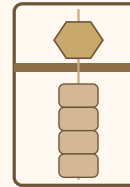
Count the objects, write the number, then show it on the abacus.



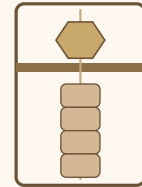
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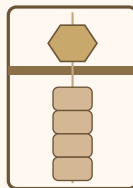
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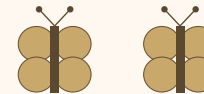
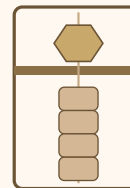
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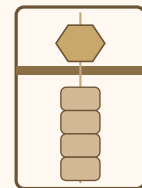
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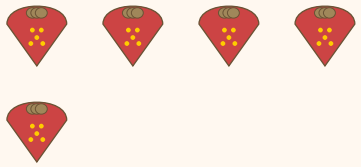


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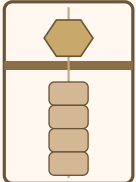


Count and Write

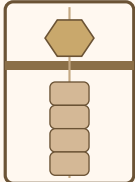
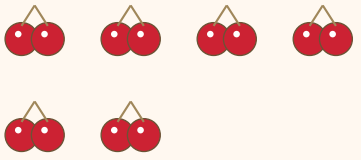
Count the objects, write the number, then show it on the abacus.

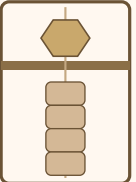
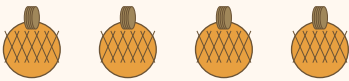
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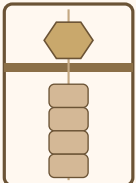
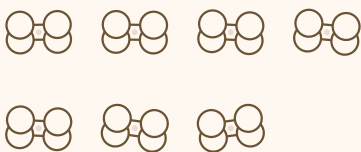

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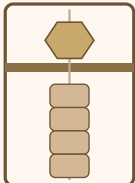
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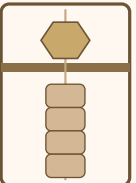
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Match the Following

Draw lines to match each number with its dot pattern and abacus.

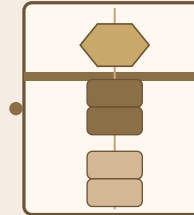
Number

Dots

Abacus

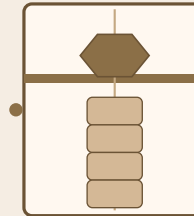
1

••••



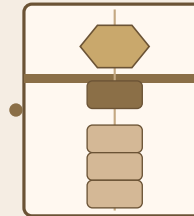
2

•••••



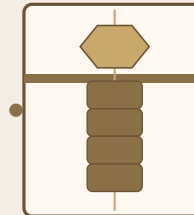
3

•••



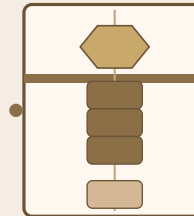
4

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5

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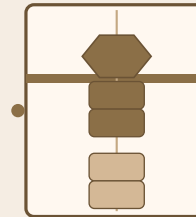


Match the Following

Draw lines to match each number with its dot pattern and abacus.

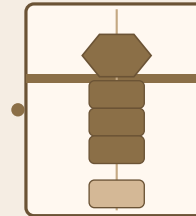
6

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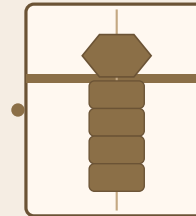
7

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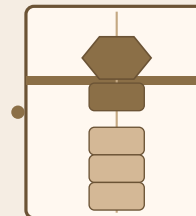
8

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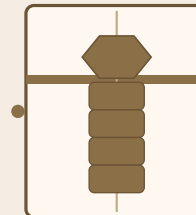
9

•••••



10

•••••



Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 1-1, 2-1, 3-1, 4-1, 2-2, 3-2

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	3	4	2
2	+1	-1	-1	-2	+2	-2	-1	+1
3	-1	+1	-1	+1	-1	+2	-2	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	1	2	4	3	2	4	1	3
2	+3	-1	-3	-1	+2	-2	+4	+1
3	-2	+3	+1	+2	-3	-1	-3	-3
Ans.								

S.No.	1	2	3	4
1	4	2	3	1
2	-1	+1	-2	+2
3	-2	-3	+1	-3
Ans.				

3 + 2 =

5 - 3 =

4 + 1 =

5 - 4 =

Direct Addition & Subtraction



Fingering Ex: 1+2, 2+1, 3+1, 5+1, 2+2, 5+2

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	2	3	4
2	+2	+1	+1	-2	+4	+3	+2	-3
3	-1	-2	-3	+1	-2	-4	-1	+2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	2	4	3	2	4	1	3	4
2	-1	-1	+1	+2	-2	+3	-1	-3
3	+3	-2	-2	-3	+1	-1	+2	+1
Ans.								

S.No.	1	2	3	4
1	1	3	2	4
2	+2	-2	+1	-1
3	+1	+2	-3	+1
Ans.				





$2 + 4 =$





$5 - 2 =$





$3 + 2 =$





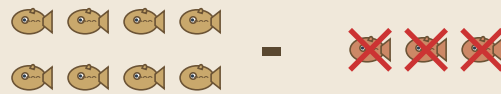
$4 - 3 =$

Direct Addition & Subtraction

Count the objects and write the answer.



$5 + 3 =$



$8 - 3 =$



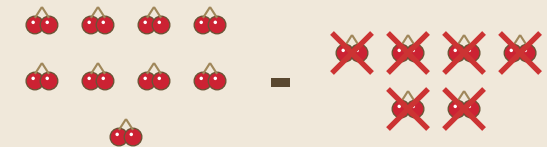
$7 - 5 =$



$4 + 5 =$



$5 + 4 =$



$9 - 6 =$

Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 1+5, 2+5, 3+5, 4+5, 5-5, 6-5

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	2	3	4
2	+6	+5	+5	+5	+7	+6	+5	+5
3	-2	-3	-4	-6	-5	-1	-1	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	7	4	5	6	8	7	5	9
2	-5	-3	-2	-1	-5	-3	+3	-5
3	+3	+6	+4	-3	+1	-2	-6	-1
Ans.								

S.No.	1	2	3	4
1	1	7	3	8
2	+5	-2	+5	-5
3	-3	+1	-6	+2
Ans.				





$1 + 5 =$





$4 - 2 =$





$2 + 3 =$





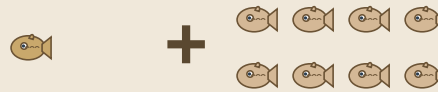
$5 - 1 =$

Direct Addition

Count the objects and write the answer.



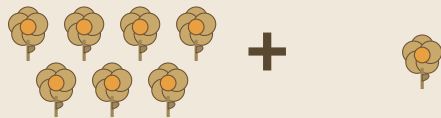
$2 + 6 =$



$1 + 8 =$



$2 + 7 =$



$7 + 1 =$



$3 + 5 =$



$4 + 4 =$



Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 1+6, 2+6, 3+6, 1+7, 2+7, 1+8

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	5	6	7
2	+6	+7	-5	-2	+7	-3	-2	-5
3	-5	-2	+7	+5	-2	+1	-3	+1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	8	7	6	9
2	+5	+8	-5	-3	-6	-5	-3	-1
3	+3	-6	+6	+5	+3	+4	+1	-8
Ans.								

S.No.	1	2	3	4
1	1	2	7	6
2	+7	+8	-5	-5
3	-5	-3	+1	+1
Ans.				





$3 + 3 =$





$5 - 4 =$





$1 + 4 =$





$4 - 2 =$

Abacus Calculation: 1 Digit 3 Rows



S.No.	1	2	3	4	5	6	7	8
1	3	7	5	2	1	4	6	8
2	+4	-2	-3	+7	+9	-2	-3	-5
3	-5	+4	+6	-5	-7	+5	-2	+3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	4	9	2	5	7	8	3	1
2	+6	-4	-1	+3	-3	-6	+6	+8
3	-7	+1	+8	-6	+5	+4	-5	-7
Ans.								

*

S.No.	1	2	3	4
1	6	9	4	7
2	-2	-7	+5	-1
3	+1	+2	-3	-5
Ans.				













$4 + 2 =$

$5 - 3 =$

$2 + 3 =$

$3 - 1 =$

Abacus Calculation: 1 Digit 3 Rows



S.No.	1	2	3	4	5	6	7	8
1	4	8	6	3	7	1	5	2
2	-1	+1	-2	-3	-5	+8	+4	-1
3	-2	-7	+5	+9	+2	-6	-7	+6
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	8	6	3	9	4	7	2	5
2	-6	-1	+7	-2	+5	-3	+6	-4
3	+7	-3	-4	-5	-8	+1	-5	+3
Ans.								

S.No.	1	2	3	4
1	9	7	4	8
2	-2	-1	-3	-5
3	+1	-5	+5	-2
Ans.				



+

$3 + 1 =$



-

$4 - 1 =$



+

$2 + 4 =$



-

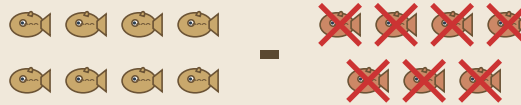
$5 - 2 =$

Direct Subtraction

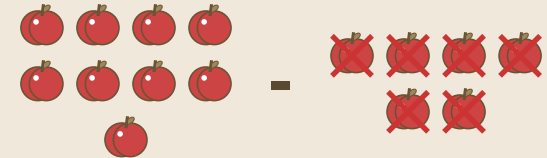
Count the objects and write the answer.



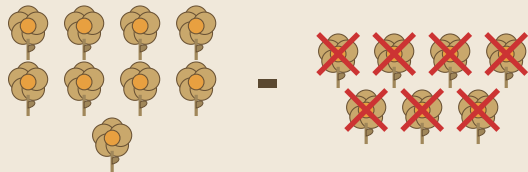
$8 - 6 =$



$8 - 7 =$



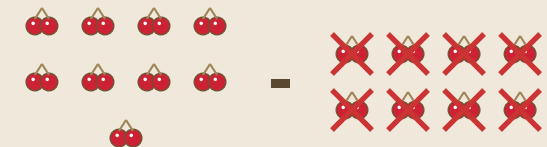
$9 - 6 =$



$9 - 7 =$



$7 - 6 =$



$9 - 8 =$



Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 6-6, 7-6, 8-6, 9-6

S.No.	1	2	3	4	5	6	7	8
1	7	8	9	6	5	4	3	2
2	-6	-6	-6	-6	+3	+6	-2	+7
3	+1	+3	-2	+7	-6	-8	+8	-6
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	6	7	8	9	6	5	4	3
2	+1	-3	-6	-6	+3	-4	+6	+7
3	-6	+5	+4	-1	-7	+8	-9	-6
Ans.								

S.No.	1	2	3	4
1	7	9	8	6
2	-6	-6	-6	-6
3	+6	+3	-1	+4
Ans.				





 $4 + 1 =$





 $5 - 2 =$





 $1 + 3 =$





 $4 - 3 =$

Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 7-7, 8-7, 9-7, 8-8, 9-8, 9-9

S.No.	1	2	3	4	5	6	7	8
1	7	8	9	8	9	9	7	8
2	-7	-7	-7	-8	-8	-9	-7	-8
3	+9	+4	-1	+5	+3	+8	+6	+9
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	9	7	8	9	8	9	7	8
2	-9	-7	-8	-7	-7	-9	-6	-7
3	+7	+8	+6	+4	+6	+5	+8	-1
Ans.								

S.No.	1	2	3	4
1	9	7	8	9
2	-8	-7	-8	-9
3	+7	+9	+5	+8
Ans.				





$2 + 2 =$





$4 - 3 =$





$3 + 3 =$





$5 - 4 =$

Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 8-8, 9-8

S.No.	1	2	3	4	5	6	7	8
1	1	7	8	4	9	2	6	3
2	+9	-8	-8	-8	-8	+8	-8	-8
3	-8	+9	+5	+9	+3	-9	+7	+8
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	9	7	8	5	3	1	4	9
2	-8	+2	-7	-8	+6	+8	-3	-8
3	+4	-8	+4	+9	-8	-7	+8	+6
Ans.								

S.No.	1	2	3	4
1	8	9	7	8
2	-8	-8	-8	-8
3	+7	+4	+9	+1
Ans.				




$5 + 1 =$




$3 - 2 =$




$4 + 2 =$




$5 - 3 =$

Abacus Calculation: 1 Digit 3 Rows



Fingering Ex: 8-8, 9-8, 9-9

S.No.	1	2	3	4	5	6	7	8
1	9	8	7	9	8	7	9	8
2	-9	-8	-7	-8	-7	-6	-9	-8
3	+7	+9	+6	+4	+3	+5	+1	+2
Ans.								



S.No.	1	2	3	4	5	6	7	8
1	9	8	7	9	8	9	7	8
2	-7	-8	-6	-9	-7	-8	-6	-6
3	+4	+9	+8	+7	+6	+3	+5	-1
Ans.								



S.No.	1	2	3	4
1	9	8	7	9
2	-7	-8	-6	-9
3	+4	+9	+8	+7
Ans.				



 + 

1 + 4 =

 - 

5 - 1 =

 + 

3 + 2 =

 - 

4 - 2 =



Composition of 5

Understanding how numbers make 5

5 can be split into two parts in different ways!

5

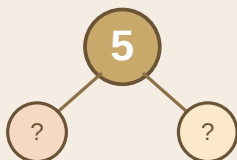
is made of two parts

4 & 1  = 5

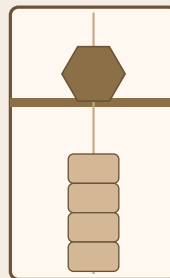
3 & 2  = 5

2 & 3  = 5

1 & 4  = 5

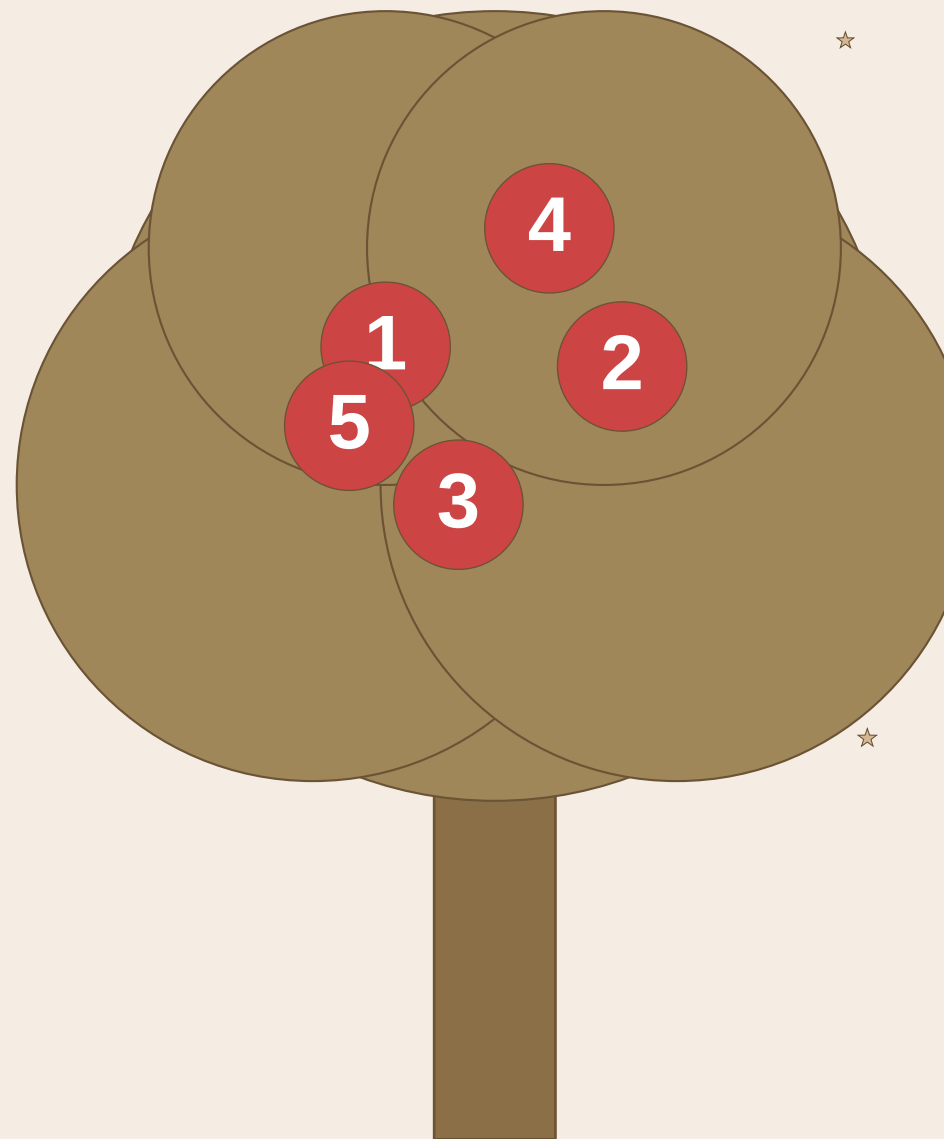


Number bond: split 5 into two parts



5

On the abacus,
5 is one upper bead.



Five apples on the tree!



Fill in the missing number:

4 + ____ = 5

____ + 3 = 5

1 + ____ = 5

____ + 2 = 5

3 + ____ = 5

____ + 4 = 5

Composition of 5

Showing combinations on the abacus

See how we make 5 on the abacus!



2 + 3 = 5

2

5

4 + 1 = 5

4

5

1 + 4 = 5

1

5

3 + 2 = 5

3

5



Abacus Calculation: Composition of 5 (+4)



+4=+5-1 | Fingering: 1+4, 2+4, 3+4, 4+4

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	2	3	4
2	+4	+4	+4	+4	+4	+4	+4	+4
3	-3	-5	-6	-7	+2	-1	-5	-3
* Ans.								

S.No.	1	2	3	4	5	6	7	8
1	2	1	3	4	1	2	3	4
2	+4	+4	+4	+4	+4	+4	+4	+4
3	-3	-5	+1	-6	-4	-5	-2	-1
Ans.								

S.No.	1	2	3	4
1	1	4	2	3
2	+4	+4	+4	+4
3	-3	-5	-6	+1
Ans.				





$2 + 3 =$





$5 - 3 =$





$4 + 1 =$





$3 - 2 =$

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	1	2	3	7	4	5	8	1
2	+3	+6	-1	-5	+2	-2	-6	-5
3	-2	-5	+4	+2	-3	+4	+3	+9
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	2	7	1	4	8	5	9	7
2	+4	-5	+3	-2	-3	+4	-6	+2
3	-3	+6	+5	+6	+1	-7	+2	-8
Ans.								

S.No.	1	2	3	4
1	3	2	6	5
2	+4	-1	+3	+2
3	-5	+7	-8	-4
Ans.				





3 + 2 =





4 - 1 =





2 + 3 =





5 - 2 =

Abacus Calculation: Composition of 5 (+3)



+3=+5-2 | Fingering: 2+3, 3+3, 4+3

S.No.	1	2	3	4	5	6	7	8
1	2	3	4	2	3	4	2	3
2	+3	+3	+3	+3	+3	+3	+3	+3
3	-4	-5	-6	+1	-1	-5	-3	-4
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	4	2	3	4	2	3	4	2
2	+3	+3	+3	+3	+3	+3	+3	+3
3	-2	-5	+1	-7	-4	-6	-3	-1
Ans.								

S.No.	1	2	3	4
1	3	4	2	3
2	+3	+3	+3	+3
3	-5	-4	-3	+2
Ans.				

+

4 + 2 =

-

5 - 4 =

+

2 + 4 =

-

4 - 1 =

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	3	2	4	7	5	8	1	6
2	+4	+6	-2	-3	+3	-5	+4	-4
3	-5	-3	+5	+1	-6	+2	-3	+3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	2	4	5	7	1	8	3	9
2	+3	-3	-5	-2	+6	-3	+5	-7
3	-4	+6	+3	-4	-5	-4	-7	+2
Ans.								

S.No.	1	2	3	4
1	4	6	5	7
2	+3	-2	-3	-5
3	-5	+3	+2	+4
Ans.				





3 + 3 =





5 - 2 =





1 + 5 =





4 - 3 =

Count and Write



Count the objects on each side, then write the addition and answer.



$4 + 2 =$



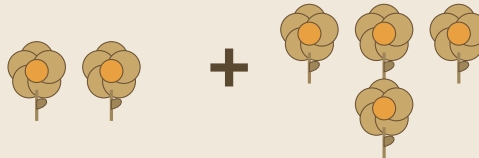
$3 + 2 =$



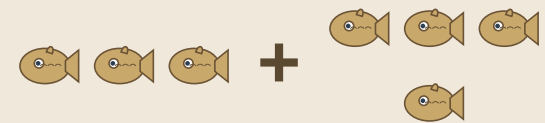
$2 + 3 =$



$4 + 1 =$



$2 + 4 =$



$3 + 4 =$



Abacus Calculation: Composition of 5 (+2)



+2=+5-3 | Fingering: 3+2, 4+2

S.No.	1	2	3	4	5	6	7	8
1	3	4	3	4	3	4	3	4
2	+2	+2	+2	+2	+2	+2	+2	+2
3	-4	-5	-3	-6	+1	-4	-2	-1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	3	4	3	4	3	4	3	4
2	+2	+2	+2	+2	+2	+2	-2	+2
3	-5	-3	+2	-6	-4	-1	+4	-7
Ans.								

S.No.	1	2	3	4
1	3	4	3	4
2	+2	+2	+2	+2
3	-4	-6	-5	+1
Ans.				


 +
 

 $2 + 2 =$


 -
 

 $4 - 2 =$


 +
 

 $3 + 1 =$


 -
 

 $5 - 3 =$

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	2	4	5	7	3	6	1	8
2	+8	+3	+2	-3	+4	-5	+7	-4
3	-3	-5	-6	+1	-2	+4	-3	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	4	3	6	9	2	5	8	1
2	+2	+5	-4	-5	+7	-3	-2	+4
3	-3	-7	+2	-3	-1	+4	-5	+3
Ans.								

S.No.	1	2	3	4
1	4	7	5	3
2	+3	-5	+2	+4
3	-2	+3	-4	-6
Ans.				





$5 + 1 =$





$3 - 1 =$





$2 + 3 =$





$4 - 2 =$

Abacus Calculation: Composition of 5 (+1)



+1=+5-4 | Fingering: 4+1

S.No.	1	2	3	4	5	6	7	8
1	4	4	4	4	4	4	4	4
2	+1	+1	+1	+1	+1	+1	+1	+1
3	-3	-5	+2	-4	-2	-6	+3	-1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	4	4	4	4	4	4	4	4
2	+1	+1	+1	+1	+1	+1	+1	+1
3	-5	+6	-3	-5	-2	-4	+2	-5
Ans.								

S.No.	1	2	3	4
1	4	4	4	4
2	+1	+1	+1	+1
3	-3	-5	-4	+2
Ans.				


 +
 

$1 + 3 =$


 -
 

$4 - 1 =$


 +
 

$3 + 2 =$


 -
 

$5 - 2 =$

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8	9	10
1	3	7	5	8	2	6	9	4	1	3
2	-4	-2	-5	-6	+4	-3	-7	+5	+7	+6
3	+6	+3	+4	+3	-5	+1	+3	-8	-3	-5
Ans.										



S.No.	1	2	3	4	5	6	7	8	9	10
1	5	8	6	3	4	7	2	9	1	5
2	+4	-3	-5	+7	-2	-6	+5	-4	+8	-2
3	-7	+2	+4	-9	+5	+8	-3	-3	-6	+4
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
1	4	7	2	8	5	3	6	9	1	4
2	+6	-5	+4	-3	-4	+7	-2	-5	+3	+5
3	-8	+3	-3	+1	+6	-8	+4	-2	-4	-7
Ans.										







 $4 + 2 =$





 $5 - 3 =$





 $2 + 2 =$





 $3 - 1 =$

Abacus Calculation: Revision

All Aboard the Speedy Scholars Express!



Solve each problem and write answers in the wheel circles!

Locomotive	Car 1	Car 2	Car 3	Car 4
SPEEDY SCHOLARS	3	1	2	4
+4	+4	+3	+2	+1
+2		+4	+1	+4
	Ans	Ans	Ans	Ans